

CAPSTONE FINAL REPORT: Intelligent Healthcare Diagnostic Assistant

Project Duration: Weeks 8 - 13

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1. Executive Summary & Mission

The objective of this capstone project was to build an end-to-end AI system that integrates intelligent agents, search, probabilistic reasoning, machine learning, NLP, fuzzy logic, and planning into a unified healthcare diagnostic and recommendation platform.

Rather than relying on a single algorithm, this system utilizes a multi-agent ensemble architecture. It functions similarly to a real medical board, where multiple specialists analyze patient data, and a central agent synthesizes their outputs to formulate a final diagnosis and a step-by-step treatment plan.

2. System Architecture & PEAS Framework

The core of the system is a Model-Based, Goal-Based Agent that operates using the classic **Perceive** → **Think** → **Act** loop.

PEAS Definition:

- **Performance:** Diagnostic accuracy, patient safety, and recommendation quality.
- **Environment:** Patient symptoms, vitals, and medical history.
- **Actuators:** Diagnosis report, treatment plan, and urgency alerts.
- **Sensors:** Symptom text input, temperature readings, and heart rate.

3. Module Implementation Details

The system is divided into seven integrated modules:

3.1. Intelligent Agent (Module 1)

Acts as the central coordinator. It receives patient intake data, consults all specialized diagnostic modules concurrently, aggregates their confidence scores using a majority-vote system, and routes the final consensus to the AI Planner.

3.2. Logic & Knowledge Base (Module 2)

Implements First-Order Logic (FOL) to act as the rule-based specialist. It utilizes Forward Chaining to derive conclusions from presented symptoms and Backward Chaining to verify suspected conditions against a medical knowledge base.

3.3. Bayesian Network (Module 3)

Calculates the probabilistic likelihood of diseases under uncertainty. It applies Naïve Bayes to compute posterior probabilities, weighing the base rate of a disease in the population against the likelihood of specific observed symptoms.

3.4. ML Diagnostic Classifier (Module 4)

A supervised learning module trained on a synthetic dataset of 2,000 patient records. It evaluates features using Decision Trees, Random Forests, and Gradient Boosting, automatically selecting the most accurate model to map symptom vectors to disease labels.

3.5. Deep Neural Network (Module 5)

A multi-layer perceptron (MLP) built for deep pattern recognition. The architecture processes an 18-dimensional symptom vector through three hidden layers (128 → 64 → 32 neurons) utilizing Batch Normalization and Dropout to prevent overfitting, terminating in a Softmax classification layer.

3.6. Fuzzy Severity Assessor (Module 6)

Operates independently of diagnosis to determine case urgency (0 to 100 scale). It fuzzifies continuous inputs (Temperature, Heart Rate, and Symptom Count) into linguistic variables, applies min/max logical rules, and uses centroid defuzzification to output severity labels ranging from LOW to CRITICAL.

3.7. AI Treatment Planner (Module 7)

Generates the final step-by-step medical plan. It models medical actions (e.g., ordering tests, administering fluids) using STRIPS planning representation and utilizes Breadth-First Search (BFS) to chart a valid path from the patient's initial state to a treated, stable goal state.

4. System Evaluation & Quality Assurance

To audit the system's performance, the evaluation module generated comprehensive metrics comparing the AI predictions against known ground-truth labels.

- **Overall Accuracy:** The Deep Neural Network achieved a peak validation accuracy of 91.00%, while the Gradient Boosting classifier achieved over 99% on the synthetic cross-validation dataset.
- **Ensemble Reliability:** Testing revealed the strength of the multi-agent approach. For instance, in complex edge cases (e.g., Meningitis presenting with generalized fever), the system successfully ignored the Bayesian Network's incorrect hypothesis by favoring the highly confident outputs of the DNN and ML classifiers.
- **Dynamic Planning:** The AI Planner successfully navigated state-space barriers, correctly routing CRITICAL patients to the ICU while assigning LOW urgency patients to home monitoring.

Metrics Tracked:

- **Accuracy:** Overall correctness (Correct / Total).
- **Precision:** How often positive predictions were right ($TP / (TP + FP)$).
- **Recall:** How often actual positives were caught ($TP / (TP + FN)$).
- **F1-Score:** The harmonic balance of precision and recall ($2 \times (P \times R) / (P + R)$).

5. Conclusion

The Intelligent Healthcare Diagnostic Assistant successfully fulfills all capstone requirements. By combining classical search and logic algorithms with modern deep learning and fuzzy control systems, the project demonstrates a robust, fault-tolerant approach to artificial intelligence in medical diagnostics.